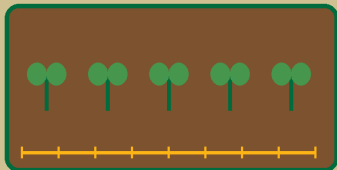


Identify System Components



IDENTIFY SYSTEM COMPONENTS

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-291 • Identify System Components

INVESTIGATION

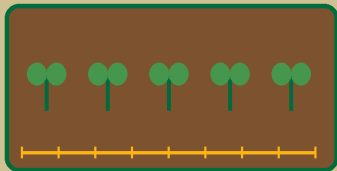
Inspect the system carefully, identify the requested feature or variable, and explain why it matters to the investigation.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Compare Growing Media



COMPARE GROWING MEDIA

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

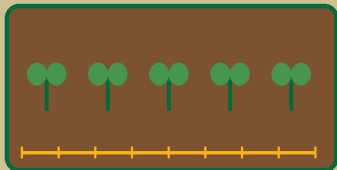
Use the same measurement or observation method for both cases.
Record the difference and identify evidence that explains it.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Measure Reservoir Volume



MEASURE RESERVOIR VOLUME

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-293 • Measure Reservoir Volume

INVESTIGATION

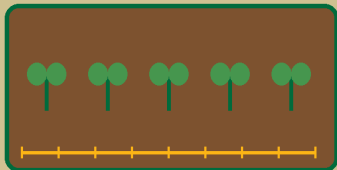
Use the appropriate teacher-approved instrument. Take measurements consistently, include units, and record the result.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Observe Root Condition



OBSERVE ROOT CONDITION

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

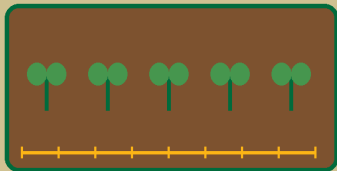
Carry out the investigation using a consistent method. Record observations or measurements and explain what the evidence indicates.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Measure pH



MEASURE PH

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

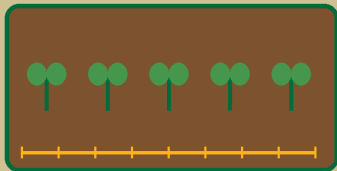
Use the appropriate teacher-approved instrument. Take measurements consistently, include units, and record the result.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Measure Electrical Conductivity



MEASURE ELECTRICAL CONDUCTIVITY

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-296 • Measure Electrical Conductivity

INVESTIGATION

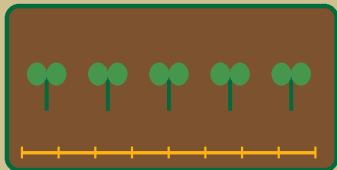
Use the appropriate teacher-approved instrument. Take measurements consistently, include units, and record the result.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Compare Water Delivery



COMPARE WATER DELIVERY

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-297 • Compare Water Delivery

INVESTIGATION

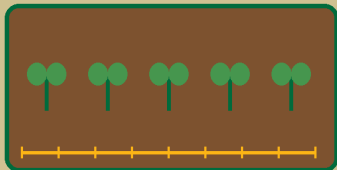
Use the same measurement or observation method for both cases.
Record the difference and identify evidence that explains it.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Check Drainage



CHECK DRAINAGE

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-298 • Check Drainage

INVESTIGATION

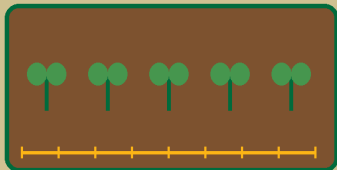
Carry out the investigation using a consistent method. Record observations or measurements and explain what the evidence indicates.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Compare Soil and Soilless Growth



COMPARE SOIL AND SOILLESS GROWTH

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-299 • Compare Soil and Soilless Growth

INVESTIGATION

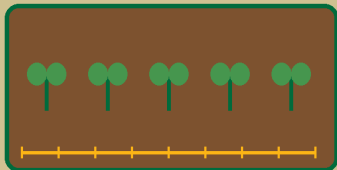
Use the same measurement or observation method for both cases.
Record the difference and identify evidence that explains it.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Evaluate the Soilless System



EVALUATE THE SOILLESS SYSTEM

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-300 • Evaluate the Soilless System

INVESTIGATION

Use at least two recorded indicators to judge performance. State one strength, one limitation, and one practical improvement.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN